

How to Falsify TNA with \$200 Hardware — and Why It Matters for Education

Introduction to TNA: A Simple Framework for Understanding System Dynamics

Dear Teacher,

Imagine explaining to your students why a river floods, a battery overflows, or why too much stress can make someone sick—all without diving into complex math. That's the power of Throughput-Based Analysis (TNA), a straightforward framework I developed to model how systems handle incoming "stuff" (like energy, waste, or costs) and what happens when they can't process it fast enough.

TNA boils down to this: Every system—whether a cell, an ecosystem, an economy, or even a population—has inflows (G , things coming in) and outflows (Ψ , capacity to handle or discharge them). If inflows exceed outflows for too long, "incoherence" (N , buildup) accumulates until the system hits a breaking point. It's like your body: if you eat but don't eliminate waste, you get sick. TNA uses simple knobs and circuits to make this visible, turning abstract ideas into hands-on experiments.

Why bring this to your classroom? TNA isn't in textbooks yet—it's a new tool—but it's perfect for secondary physics or science. With a \$200 kit, students twist dials to see collapse in real time: no equations needed, just observation and discussion. It bridges subjects—physics (energy flow), biology (homeostasis), economics (supply-demand imbalance)—and teaches critical thinking: "What if we increase capacity? Does it prevent failure?" It's falsifiable (build it, test it, prove it wrong if you can), engaging, and low-cost. Any advanced electronics student can help assemble the device during practical classes, turning it into a collaborative project that reinforces circuit skills while exploring real-world systems. Request one for your lab, and watch students grasp why systems fail mechanically, not magically. It's not ideology—it's physics made practical.

The core differential expression of TNA dynamics is:

$$\frac{dN}{dt} = G(t) - \Psi(t)$$

where:

- $G(t)$ = inflow or disorder generation
- $\Psi(t)$ = discharge or system-capacity
- $N(t)$ = accumulated load

If $G > \Psi$ for a sustained period, $N(t)$ crosses a threshold and the system collapses.

If a physical implementation fails to show collapse under $G > \Psi$, the model would be falsified.

1. A \$100–200 Analog Hardware Test

A minimal analog-computer setup can be built with consumer components:

- **Arduino Uno or Raspberry Pi Zero (\$10–30):** optional data logging and threshold triggers.
- **Basic op-amps (LM358 or similar) (\$5–10):** integrators and summing amplifiers.
- **Resistors, capacitors, breadboard (\$10–20):** RC accumulation acts as $N(t)$.
- **Sensors, small PSU, multimeter (\$20–50).**
- **LED or buzzer trigger (\$5)** to mark collapse when N exceeds a comparator threshold.

Total cost: **\$100–200**, depending on tools already available.

Complete “DIY analog computer” kits online often fall in the \$50–100 range.

2. Physical Mapping of the Equation

- An **op-amp integrator + capacitor** stores charge: this is $N(t)$.
- A **constant voltage source** represents $G(t)$.
- A **variable resistor or adjustable sink** represents $\Psi(t)$.
- A **comparator** triggers an LED/buzzer at $N > N_{\max}$.

If $G = 2 \text{ V/s}$ and $\Psi = 1 \text{ V/s}$, voltage rises linearly and crosses threshold in predictable time.

Measured collapse = model validated.

Absence of collapse = model falsified.

No appeals to sociology or interpretation — just **instrumentation**.

3. Why It Is Falsifiable

A critic can vary:

- inflow amplitude
- discharge resistance
- threshold values
- noise injection
- non-linear sinks

If the system **fails to collapse when** $G > \Psi$, the model is refuted in physical analogy.
If collapse occurs reliably, the mechanism is **reproducible**.

That is the essence of scientific falsification.

4. Why Schools Should Use It

A \$200 analog testbed turns abstract differential equations into **visible dynamics**:

- adjust one knob (inflow G)
- adjust another knob (capacity Ψ)
- watch a capacitor voltage ramp until it trips

Students understand in minutes what textbooks explain in weeks: **accumulation and threshold failure**.

This is immediately transferable across disciplines:

Biology

Toxic metabolic by-products vs. excretion.

If elimination capacity drops, the organism dies.

Ecology

Predator–prey feedback, or invasive growth exceeding environmental absorption.

Economics

Monetary injection exceeding productive throughput → inflation.

Demography

Rising reproductive cost vs. institutional support → falling birth rates.

Climate

CO₂ accumulation vs. natural sinks.

This is not ideological. It is **systems dynamics made tangible**.

5. Why It Outperforms Pure Mathematics in Pedagogy

For secondary-level education:

- no differential-equation background required
- immediate causal understanding
- replicable classroom experiments
- bridges physics, biology, economics, and ecology
- cultivates falsifiability, not faith

A teacher could stock **one kit per classroom for less than a projector**.

In practical learning environments, nothing teaches entropy, accumulation, and capacity constraints faster.

6. Educational Payoff

Students would learn:

- threshold systems
- non-linear collapse
- steady-state equilibria
- feedback vs. failure
- open vs. closed systems
- why “constraints” are physical, not political

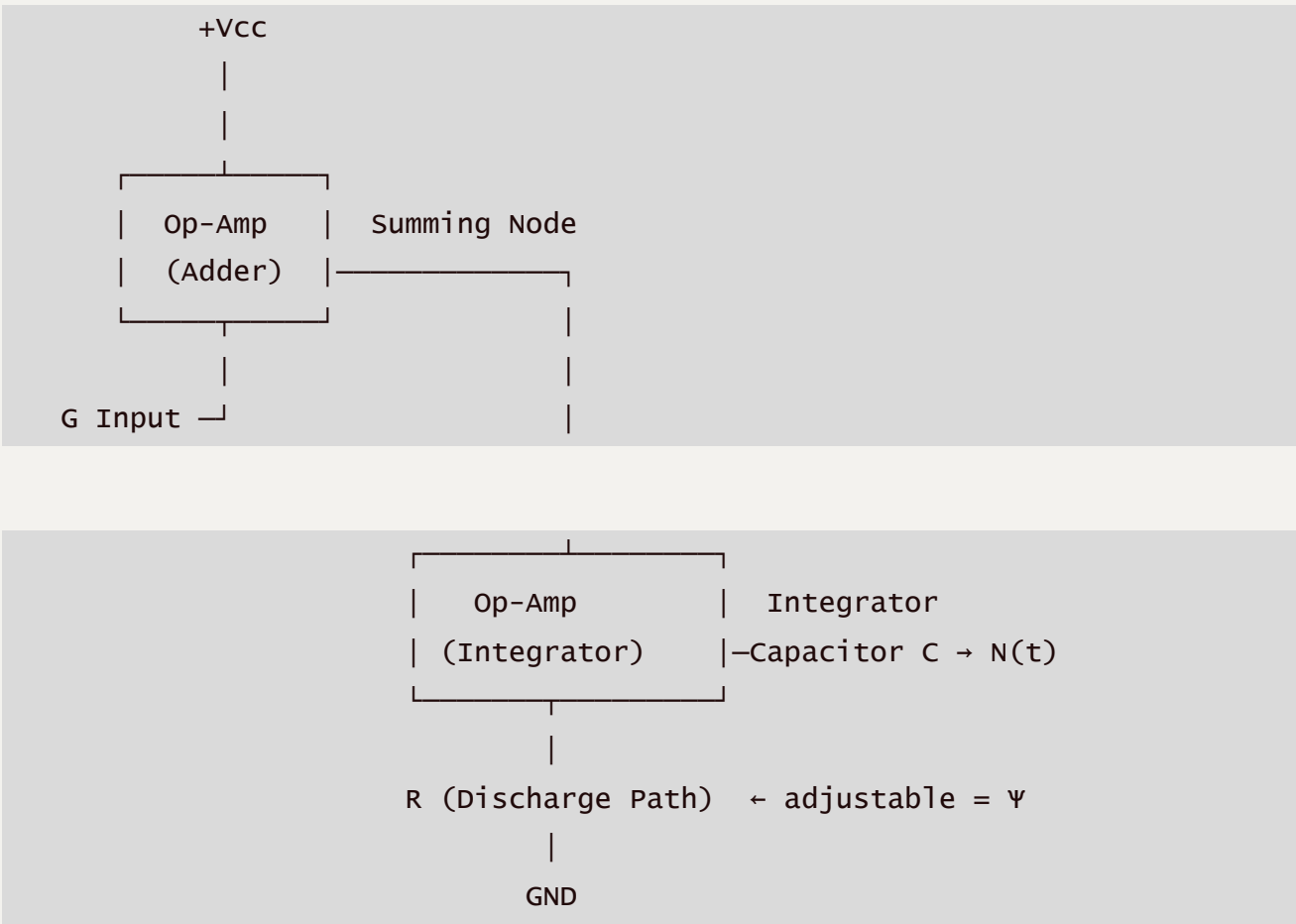
It is an accessible on-ramp to **systems engineering literacy**, not ideological persuasion.

Appendix A — Analog-Computer Schematic for TNA Collapse Equation

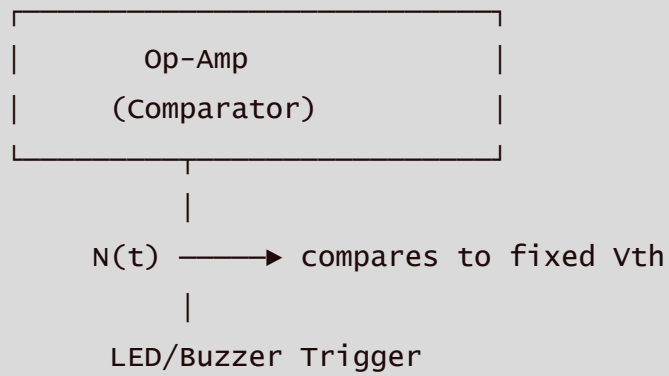
This schematic implements the differential equation

$$\frac{dN}{dt} = G - \Psi$$

using one op-amp summing stage, one op-amp integrator, and a comparator for overflow.



Comparator Stage for Collapse Threshold



How it works

- G = fixed reference voltage, e.g., 1–3 V.
- Ψ = adjustable resistance controlling discharge.
- The integrator capacitor stores charge as a physical analog of accumulation $N(t)$.
- When the comparator detects $N(t) > V_{th}$ (e.g., 5 V), it activates a visible alarm — marking **collapse**.
- If turning Ψ low prevents collapse, the equation is falsified.

Appendix B — Bill of Materials (< \$200 USD total)

COMPONENT	QTY	APPROX. COST	NOTES
Arduino Uno or Raspberry Pi Zero	1	\$10–30	Optional; used for datalogging
LM358 Dual Op-Amp (or TL082)	2–3	\$5–10	1 adder, 1 integrator, 1 comparator
Breadboard + jumper wires	1 kit	\$10–15	Universal prototyping
Assorted resistors (1 k Ω –1 M Ω)	1 pack	\$5–8	Ψ is tuned via resistance
Assorted capacitors (10 μ F–470 μ F)	1 pack	\$5–10	Selected for integration constant
Variable potentiometer (1 k Ω –100 k Ω)	1–2	\$2–6	Used as discharge valve (Ψ)
0–12 V DC supply or USB-lab PSU	1	\$10–20	Stable analog powering
Digital multimeter	1	\$15–25	Or built-in Arduino ADC
LED + 220 Ω resistor	1–2 sets	\$1–2	Visual collapse indicator
Piezo buzzer (5 V)	1	\$2–4	Optional audible alarm
USB logic analyzer or oscilloscope	1	\$20–80	For plotting $N(t)$ over time
Sensor wires / clips	1 set	\$5	Safety and convenience
Protective case	1	\$5–15	Classroom durability

Total:

90*low* — *end*—**180 high-end**, depending on PSU and oscilloscope choice.

No item requires special sourcing. All parts are available from:

- generic Amazon kits
- AliExpress
- local electronics suppliers
- “Arduino starter” bundles

Optional Enhancements

A. Data Logging

- Arduino reads capacitor voltage (A0 pin).
- Log every 50–100 ms.
- Export CSV to plot collapse curves.

B. Threshold Variations

- Replace comparator threshold with a potentiometer — explores **different collapse definitions**.

C. Non-Linear Ψ

- Add a diode network to simulate **institutional saturation** (discharge declines at high load).
- Test if $G > \Psi(N)$ accelerates collapse.

D. Noise Injection

- Add small white-noise signal into G input.
- Test robustness of collapse prediction.

Experiment 1 — Fixed Ψ , Increase G Until Collapse (“Overflow Test”)

Goal

Show that when **input exceeds discharge**, accumulation rises and triggers collapse — the physical analog of

$$\frac{dN}{dt} = G - \Psi > 0 \Rightarrow N(t) \rightarrow \text{overflow}.$$

Setup

- Set the discharge potentiometer (Ψ) to a mid-range resistance (e.g., $10\text{k}\Omega$).
- Set comparator threshold $V_{th} = 5\text{ V}$.
- Feed a small constant input $G = 1.0\text{ V}$.

Steps

1. Power the circuit.
2. Measure capacitor voltage every 2 seconds.
3. Increase G gradually ($1.0\text{V} \rightarrow 1.5\text{V} \rightarrow 2.0\text{V} \rightarrow 3.0\text{V}$).
4. When the LED/buzzer fires, record the time.

Expected Observations

- For $G \approx \Psi$, the capacitor voltage stabilizes.
- Once $G > \Psi$, voltage ramps until threshold trips.
- Plot $N(t)$: students see a **linear or exponential charge curve**, depending on RC choice.

Lesson

No ideology required — if a system cannot discharge what enters, collapse is mechanical.

Experiment 2 — Fix G , Tune Ψ to Prevent Collapse (“Institutional Rescue”)

Goal

Show that “capacity” (Ψ) prevents collapse if increased above a critical bound:

$$\Psi > G \Rightarrow \frac{dN}{dt} < 0.$$

Setup

- Fix G at a high constant level (e.g., 3.0 V) that previously caused overflow.
- Start Ψ at high resistance (weak discharge).

Steps

1. Observe capacitor voltage rising toward overflow.
2. Now lower resistance (turn the potentiometer).
3. Identify the exact Ψ at which the voltage stabilizes.
4. Log the equilibrium voltage.

Expected Observations

- With weak Ψ , collapse repeats.
- At a narrow critical setting, the system stabilizes.
- Students witness a **phase transition**: one twist of a knob flips the sign of the derivative.

Lesson

The physical system displays a **threshold inequality** — the analog of preventing inflation, demographic collapse, or reactor meltdown.

Experiment 3 — Move the Threshold and Redefine “Collapse” (“Policy Test”)

Goal

Demonstrate that collapse depends on **where you draw the safety line**, not on opinion — a direct analogy to macroprudential buffers, hospital triage saturation, or ECB inflation targets.

$$N(t) > N_{\max} \Rightarrow \text{failure signal.}$$

Setup

- Pick a single $G = 2.0$ V and weak Ψ so the system climbs.
- Replace the comparator’s fixed reference with a potentiometer.

Steps

1. Run the experiment until $N(t)$ rises.
2. Slowly lower the threshold (e.g., from 5V down to 3.5V).
3. Observe the earlier activation of collapse alarm.
4. Raise threshold (up to 6V) and see delayed alarm.

Expected Observations

- Collapse appears sooner when the safety rule is strict.
- Collapse appears later when the rule is lax.
- Students learn how **risk tolerance shifts outcomes**, even with identical physics.

Lesson

Policy is just **moving a comparator voltage** — not magical ideology.

What these three experiments teach (in 6 short sentences)

- Systems collapse when inflow > outflow.
- Stabilization requires raising discharge capacity.
- Safety depends on thresholds, not hopes.

- “Equilibrium” is a knife-edge condition.
- Collapse is predictable long before it happens.
- Anyone can falsify the model by building a counterexample.